

Empirical Assessment of Storm-time Thermospheric Density Inversion Methods from LEO POD Data

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Key Points:

- Orbit-effective densities from energy dissipation rate and POD-accelerometry match accelerometer truth with r-squared above 0.97.
- At sub-orbital resolution both methods achieve pooled r-squared above 0.92 against native accelerometer densities but uncertainty roughly doubles to triples.
- Retrieval error falls below 10 percent where drag exceeds $4\text{e-}7\text{ m/s}^2$ but grows above 30 percent below $1\text{e-}8\text{ m/s}^2$.

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Abstract

Thermospheric mass density represents one of the largest sources of operational uncertainty for spacecraft in Low Earth Orbit (LEO), particularly during geomagnetic storms. The growing population of Global Navigation Satellite System (GNSS)-equipped LEO satellites presents a dataset of opportunity: Precise Orbit Determination (POD) data streams can be used to estimate thermospheric density along their paths. We present a multi-timescale benchmark of the two main cooperative density inversion methods, Energy Dissipation Rate (EDR) and POD-accelerometry (POD-A), against accelerometer-derived effective densities over 49 geomagnetic storms spanning CHAMP and GRACE-FO-A. At orbit-effective cadence, both methods achieve $r^2 \geq 0.97$ and $\text{SD}\% \leq 22\%$, with EDR outperforming POD-A on every metric and at roughly $6\times$ lower computational cost. Increasing the fit-span to three orbits reduces $\text{SD}\%$ to $\sim 13\%$ when compared against accelerometer effective densities computed over the same interval, but r^2 against native-cadence accelerometer data drops sharply beyond ~ 45 minutes. Sub-orbital arcs of 30–45 minutes offer the best compromise, achieving $r^2 = 0.93\text{--}0.94$ and $\text{SD}\% \approx 42\text{--}46\%$ against native-cadence data. Retrieval accuracy depends strongly on drag-acceleration magnitude, with RMS errors below 10% where drag exceeds $4 \times 10^{-7} \text{ m s}^{-2}$ but growing above 30% below 10^{-8} m s^{-2} . These results provide empirical observation-error statistics to support quality-control thresholds and observation-operator tuning in next-generation assimilative models.

Plain Language Summary

Geomagnetic storms heat and expand Earth’s upper atmosphere, sharply increasing drag on satellites and degrading orbit predictions used for collision avoidance. Accurate density measurements are critical during storms, yet today’s best forecasting models rely on observations from few dedicated tracking objects. Thousands of satellites now broadcast precise positioning data that could be converted into density estimates. We tested two techniques for doing so (one based on orbital energy changes, the other on accelerations extracted from precise positions) against independent accelerometer measurements from two scientific satellites across 49 storms. When each method averages density over a full orbit, both agree with the reference to within about 14–21%, and averaging over longer spans tightens this to roughly 13%. However, orbit-long averages blur the rapid density swings that matter most during storms. Segments of 30–45 minutes strike the best balance, recovering fine structure while keeping errors manageable. Accuracy also depends on how much drag a satellite experiences: errors are small for high-drag orbits but grow at higher altitudes where the atmosphere is thinner. Routine sharing of precise tracking data from large satellite fleets could deliver the dense, timely observations needed to improve space-weather forecasting when it matters most.

1 Introduction

The recent proliferation of satellites in Low Earth Orbit (LEO), combined with intensifying solar activity associated with the current peak of Solar Cycle 25, is heightening the operational risks posed by mischaracterisation of thermospheric behaviour. Of particular concern is the inability of current thermospheric models to accurately resolve density variations during geomagnetic storms, which remain a major source of operational uncertainty (Mehta & Linares, 2017). This inadequacy leads to the degradation of orbital products used in space traffic management (e.g., probability of collision estimates, predicted ephemerides, state uncertainties) (Bussy-Virat et al., 2018) and can occasionally culminate in uncontrolled spacecraft re-entries (Fang et al., 2022).

Three broad families of thermospheric density model serve the operations community: semi-empirical climatologies (e.g. NRLMSISE-00, JB2008) (Picone et al., 2002; Bowman et al., 2008), first-principles general circulation models (e.g. TIE-GCM, WAM-IPE)

(Qian et al., 2014; Brown & Elvidge, 2024), and assimilative models (e.g. HASDM, AE-NeAS) that blend observations with an underlying model (Bruinsma et al., 2023; Elvidge & Angling, 2019). Improvements in all three are actively sought by the community (Mutschler, Tobiska, et al., 2023; Mehta et al., 2022), particularly during geomagnetic storms where model errors are largest. All three families are driven by space weather indices that can themselves become unreliable during extreme events (Shprits et al., 2019; Yamazaki et al., 2024), compounding the degradation in orbit propagation and conjunction assessment (Parker et al., 2024). The most accurate operational system, HASDM (Storz, 2002; Picone et al., 2005), is proprietary to the U.S. Government and relies on Energy Dissipation Rates (EDR) derived from ~ 80 uncooperatively-tracked objects in eccentric orbits (Storz, 2002). Its strong performance has motivated a growing number of assimilative frameworks that aim to ingest broader observational data sources (Sutton, 2018; Gondelach & Linares, 2020; Elvidge & Angling, 2019; Brown & Elvidge, 2024). Given the growing number of cooperatively tracked objects in LEO and the potential improvement in data quality they offer over trajectories derived from uncooperatively tracked objects, interest is growing in assimilating cooperative POD-derived densities into these models (Mutschler, Axelrad, et al., 2023; Fitzpatrick et al., 2025). If such data can be processed and ingested at scale, they are likely to enable a step change in the accuracy of assimilative models. However, more work is needed to understand how these densities should be processed, what accuracy they can achieve, and how they behave across different conditions.

The flip side of the recent increase in the number of satellites in LEO has meant that GNSS data have emerged as a potentially globally scalable, high-resolution, near-real-time set of “signals of opportunity” for thermospheric density estimation. This opportunity is materialising through a rapid growth in the pool of available POD data from both scientific and commercial missions (Arnold et al., 2023; Schreiner et al., 2022; Mutschler, Axelrad, et al., 2023; Fitzpatrick et al., 2025). If the uncooperatively-derived densities that drive HASDM can achieve such strong performance, it stands to reason that assimilative models using a greater number of cooperative observations should be able to see even greater improvements. In view of the mounting criticality of storm-time density retrieval, the latest International Space Weather Action Teams Working Meeting (Mehta et al., 2022) explicitly called for the community to explore new high-quality density data streams and ways of achieving “high-resolution monitoring of geomagnetic storms”.

Despite the growing interest in POD-derived densities, little to no systematic observational characterization of their performance under geomagnetic storm conditions has been carried out. This gap in benchmarking limits the community’s ability to make informed decisions about how best to integrate POD-derived data into model development and operational workflows. Recent simulation-based work by Ray et al. (2024) took an important step toward addressing this gap, concluding that a POD-A approach should outperform EDR-based approaches especially under lower drag scenarios (where drag acceleration $\lesssim 10^{-8} \text{ m s}^{-2}$). To date, however, no equivalent study exists using real observational data. This paper aims to fill that gap. Specifically, we address three questions:

1. How accurately do EDR and POD-A recover orbit-effective thermospheric density during geomagnetic storms, and does either method hold a systematic advantage?
2. What is the trade-off between temporal resolution and retrieval accuracy as arc length is varied?
3. What is the relationship between density retrieval accuracy and drag-acceleration magnitude, and do the observed trends corroborate predictions made in the literature?

We present a multi-timescale, storm-focused ($n = 49$ storms) benchmark of POD-derived thermospheric densities, comparing EDR and POD-A against accelerometer-derived

Table 1. Candidate LEO spacecraft with publicly available SP3 precise-orbit data. Data-access URLs: GFZ Potsdam products are available from <https://isdc.gfz-potsdam.de>; ESA Swarm data from <https://swarm-diss.eo.esa.int>; Spire and PAZ orbits from the NASA CSDA Satellite Data Explorer (<https://csdap.earthdata.nasa.gov>); COSMIC-2 from <https://data.cosmic.ucar.edu>; Planet Labs data by agreement with Planet Labs, Inc.

Spacecraft	Data provider	N_{sat}	Alt. (km)	Latency
GRACE-FO-A/B, TerraSAR-X/TanDEM-X	GFZ Potsdam	4	~490	30 min – 2 d
Swarm A/B/C	ESA Swarm	3	~460	~1 d
Planet Labs (Flock)	Planet Labs	200+	~475	Unspecified
Spire (LEMUR-2 constellation)	NASA CSDAP	100+	~400–600	N/A
PAZ (Spanish X-band SAR)	NASA CSDAP	1	~514	N/A
COSMIC-2	UCAR	6	~520–720	~1 d
Sentinel series	GFZ Potsdam	7	~700	30 min – 2 d

effective densities from CHAMP and GRACE-FO-A. At orbit-effective cadence, both methods achieve $r^2 \geq 0.97$ and $\text{SD}\% \leq 22\%$, with EDR outperforming POD-A on every metric at roughly $6\times$ lower computational cost. Increasing the fit-span to three orbits reduces $\text{SD}\%$ to $\sim 13\%$, but r^2 against native-cadence accelerometer data drops sharply beyond ~ 45 minutes. Sub-orbital arcs of 30–45 minutes offer the best compromise ($r^2 = 0.93\text{--}0.94$, $\text{SD}\% \approx 42\text{--}46\%$ against native-cadence data), providing empirical observation-error statistics to support observation-operator tuning in next-generation assimilative models.

2 Data and Methods

2.1 Spacecraft and Orbit Source selection

The sources of precise orbits that were considered are outlined in Table 1. The Rapid Science Orbits (RSO) were selected for this study (Schreiner et al., 2022). The RSOs exhibit latencies ranging from as little as 45 minutes to up to 2 days; this delivery latency determines how quickly retrieved densities become available operationally. Satellite Laser Ranging (SLR) residuals are approximately 1–2 cm for RSOs in LEO (Schreiner et al., 2022). According to Selvan et al. (2023), this level of accuracy places the RSOs within the top 20–35% of POD literature over the past two decades.

2.2 Selection of Storms

Geomagnetic storms are notoriously poorly characterized by empirical density models (Oliveira et al., 2021, 2017; Mutschler, Tobiska, et al., 2023), yet they represent periods of significant stress on satellite operations. Increasing the pool and quality of data pertaining to thermospheric behaviour during storm events is crucial for improving existing thermospheric models in two main areas (Fang et al., 2022; Berger et al., 2023): as a source of data to be assimilated within nowcast models and as a source of scientific observations to enhance understanding of thermospheric behaviour.

Different categories of geomagnetic storms (G1–G5) (NOAA Space Weather Prediction Center, 2024) trigger distinct physical phenomena in the magnetosphere-ionosphere-thermosphere system and exhibit varying signatures in density fluctuations (Knipp et al., 2021; Astafyeva et al., 2017; Laskar et al., 2023; Oliveira et al., 2017). To evaluate our method across the spectrum of storm behaviours, we selected 49 storms (22 for CHAMP, 27 for GRACE-FO-A) spanning NOAA G-scale categories G2–G5. Figure 1 places them in the context of the F10.7 solar-flux record.

The determination of the time window for storm analysis was guided by prior findings from Oliveira et al. (2021), which noted that the thermosphere typically peaks in density approximately 12 hours following Sudden Storm Commencement (SSC) and reverts to baseline levels within about 72 hours. SSC is traditionally defined by a sharp increase in the horizontal magnetic field, usually associated with a southward turning of the interplanetary magnetic field (IMF) B_z component.

Each storm analysis window is centred on the time of minimum Dst and extends from 48 hours before to 72 hours after this epoch, capturing the pre-storm baseline, the main phase, and the recovery. Dst was preferred over Kp because Kp saturates at 9 and is reported at 3-hour cadence, whereas Dst does not saturate and is available at 1-hour resolution (Yamazaki et al., 2024).

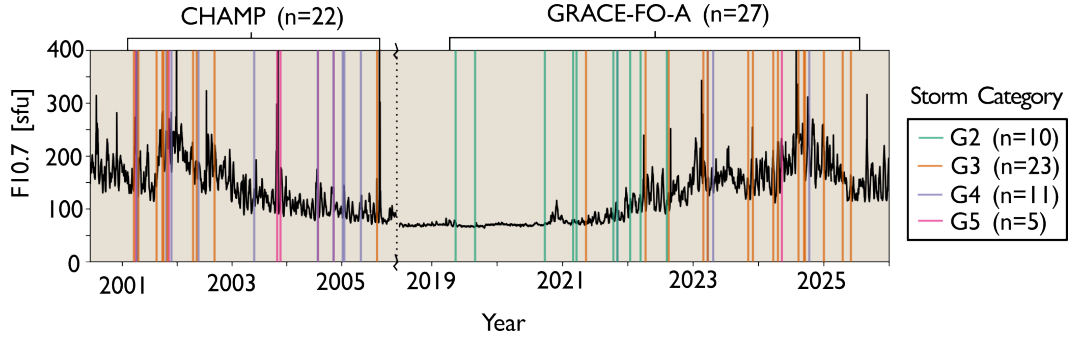


Figure 1. Timeline of the 49 selected geomagnetic storms overlaid on the daily F10.7 solar flux record. Vertical lines mark each storm, coloured by NOAA G-scale category. The CHAMP storms (2001–2005) were observed during Solar Cycle 23, while the GRACE-FO-A storms (2019–2025) fall within Solar Cycle 25.

2.3 Density Retrieval Methods

2.3.1 Effective density

Both density retrieval methods in this study produce *effective densities* in the sense of Picone et al. (2005): the velocity-weighted average that, when held constant over a given arc, reproduces the observed drag-induced energy loss. For a co-rotating atmosphere (no winds) the definition reduces to

$$\rho_{\text{eff}} = \frac{\int_{t_0}^{t_1} \rho v_{\text{rel}}^2 |\mathbf{v}| dt}{\int_{t_0}^{t_1} v_{\text{rel}}^2 |\mathbf{v}| dt} \quad (1)$$

where $\mathbf{v}$ is the inertial velocity, $\mathbf{v}_{\text{rel}} = \mathbf{v} - \boldsymbol{\omega}_{\oplus} \times \mathbf{r}$ is the velocity relative to the co-rotating atmosphere, and t_0, t_1 delimit the averaging arc. The weighting kernel $v_{\text{rel}}^2 |\mathbf{v}|$ is proportional to the instantaneous drag power; it emphasizes perigee—where the satellite is fastest and the atmosphere densest—rather than treating all epochs equally as an arithmetic mean would. This distinction is important for eccentric orbits and for any comparison with quantities derived from integrated energy loss.

Although originally formulated between successive perigee passages (Picone et al., 2005), Eq. (1) is valid for *any* choice of integration limits t_0 and t_1 . When the arc spans

one full revolution (perigee to perigee) we refer to the result as an *orbit-effective* density; when a shorter arc is used we refer to it as an *arc-effective* density. In both cases the physical quantity is the same Picone effective density of Eq. (1)—only the integration interval differs.

Orbit-effective cadence (one value per revolution) is standard in storm-time thermosphere studies (Fitzpatrick et al., 2025; Chen et al., 2014; Bruinsma et al., 2018; Li et al., 2017) and matches the resolution most commonly assimilated by contemporary physics-based filters (Matsuo et al., 2012; Sutton, 2018; Sutton et al., 2021). However, next-generation data-assimilation systems are expected to ingest density at sub-orbital cadences, and the feasibility of doing so depends on the drag signal strength—which increases substantially during geomagnetic storms. We therefore consider both orbit-effective and arc-effective resolutions in this study, using Eq. (1) throughout.

The TU Delft accelerometer-derived densities (Siemes et al., 2023), which serve as the reference truth in this study, are provided as instantaneous values at 10-second cadence. To ensure a like-for-like comparison, we compute effective densities from the accelerometer time series by applying Eq. (1) over the same arc used by the retrieval method under evaluation. This guarantees that all quantities being compared represent the same physical average over the same time interval.

The following density retrieval approaches are considered:

1. **POD-Accelerometry** (POD-A; Section 2.3.2): Satellite accelerations are obtained by numerical differentiation of the precise orbit velocities. Conservative and major non-conservative forces are modelled and subtracted at each time step, yielding a pointwise drag-acceleration time series. Effective density is then obtained by integrating this series with the Picone weighting of Eq. (1) over the chosen arc.
2. **Energy Dissipation Rate** (EDR; Section 2.3.4): The change in orbital energy over each arc is computed from the ephemeris velocities and positions—without numerical differentiation. Gravitational and other non-drag forces are accounted for via work integrals along the arc. The residual energy change is equated to the work done by drag, yielding the Picone effective density by construction (Sutton et al., 2021; Fitzpatrick et al., 2025).
3. **Accelerometer reference** (ACC): Orbit-effective and arc-effective accelerometer densities are computed from the native 10-second TU Delft data (Siemes et al., 2023) using Eq. (1), as described above. These data are widely recognized as a robust benchmark (March et al., 2021; Mehta et al., 2017; Gondelach & Linares, 2020; Kuang et al., 2014; Emmert, 2015).

For each storm, the effective density time series from every retrieval method is multiplicatively de-biased against the accelerometer reference using the median log-ratio:

$$\rho_{\text{method}}^*(t) = \rho_{\text{method}}(t) \exp(-\hat{\beta}), \quad \hat{\beta} = \text{median} \left\{ \ln \rho_{\text{method}}(t_i) - \ln \rho_{\text{ACC}}(t_i) \right\}_i \quad (2)$$

This multiplicative correction is appropriate because thermospheric densities are approximately log-normally distributed (Emmert, 2015; Picone et al., 2005); an additive (arithmetic) offset would under-correct at high densities and over-correct at low densities. The correction compensates for constant scaling errors arising from uncertainties in the drag coefficient (C_D), the effective cross-sectional area (A), or the spacecraft mass (m), which enter both inversions as a fixed ballistic coefficient $B = C_D A / m$. In operational settings such biases are typically absorbed by a fitted ballistic coefficient, Parker and Linares (2026) argue that this fitting is inherently circular, since the fitted B partially absorbs errors in the density model itself, and propose bypassing density inversion entirely in favour of a direct mapping from space weather drivers to satellite dynamics. Removing these biases here allows us to focus on relative storm-time variability.

These shorthands (POD-A, EDR, ACC) are used throughout the figures.

2.3.2 *POD-accelerometry density inversion*

POD-accelerometry refers to a family of methods that recover atmospheric drag from precise orbit ephemerides by isolating the drag acceleration implicit in the tracking data. Two commonly used approaches are: (i) ingesting positions and velocities into a reduced-dynamic orbit determination filter that simultaneously estimates density corrections through a stochastic process, and (ii) numerically differentiating the velocities to yield accelerations directly, from which modelled non-drag forces are subtracted to isolate the drag signal (Bezděk, 2010; Calabia et al., 2015; Calabia & Jin, 2017). Ray et al. (2024) adopt the first approach using a second-order Gauss–Markov process within an iterated extended Kalman filter-smoother.

This study follows the numerical-differentiation strategy but departs from earlier implementations in two important respects. First, whereas Calabia et al. (2015) and Bezděk (2010) obtain accelerations by double differentiation of positions, we differentiate the SP3 velocities only once. Because the velocities delivered by the POD process are determined within the orbit fit rather than derived from positions, a single differentiation amplifies high-frequency noise substantially less than the double-differentiation approach. Second, rather than applying temporal smoothing to the recovered density, we integrate the raw pointwise drag-acceleration time series with the Picone velocity weighting of Eq. (1) over a chosen arc. The integration inherently suppresses stochastic noise while preserving arc-averaged fidelity, and produces a quantity that is directly comparable to the accelerometer reference densities computed with the same weighting (Section 2.3.1).

The SP3 ephemeris provides positions and velocities at 30-second intervals. Each velocity component is interpolated to a 0.01 s cadence using a cubic spline (piecewise third-degree polynomial with not-a-knot boundary conditions; knots at each 30 s SP3 epoch). A Savitzky–Golay pre-filter (window of 21 points = 0.21 s, polynomial order 7) is applied to the interpolated velocities to suppress sub-second spline-interpolation artefacts before differentiation. Accelerations are then obtained by central finite differences at the 0.01 s step size; interpolation finer than 0.01 s was found to yield negligible improvement. The resulting acceleration time series is downsampled from 0.01 s to 15 s to reduce the computational burden of the subsequent force-model evaluations.

The total observed acceleration $\mathbf{a}_{\text{obs}}$ is decomposed as

$$\mathbf{a}_{\text{obs}} = \mathbf{a}_{\text{con}} + \mathbf{a}_{\text{drag}} + \mathbf{a}_{\text{srp}} + \mathbf{a}_{\text{erp}} + \mathbf{a}_{\text{other}} \quad (3)$$

where $\mathbf{a}_{\text{con}}$ is the gravitational acceleration (including third-body perturbations), $\mathbf{a}_{\text{srp}}$ is solar radiation pressure, $\mathbf{a}_{\text{erp}}$ is Earth radiation pressure, and $\mathbf{a}_{\text{other}}$ collects smaller effects such as thermal re-radiation and antenna thrust. These residual terms are typically $\lesssim 1\%$ of the drag acceleration, particularly during geomagnetic storms when drag dominates (Bhattarai et al., 2022; March et al., 2019; Doornbos et al., 2010), and their computation requires detailed spacecraft thermal and attitude data that are not generally available. We therefore set $\mathbf{a}_{\text{other}} = 0$.

The drag coefficient C_D and cross-sectional area A are held constant at the values listed in Table 2. In reality, C_D varies with thermospheric temperature, composition, and gas–surface interaction physics: the energy accommodation coefficient depends on atomic-oxygen abundance and incident kinetic energy, producing orbit-level C_D excursions of $\sim 10\text{--}15\%$ (Pilinski et al., 2013; Mehta et al., 2014), with larger departures possible during geomagnetic storms when temperature and composition change sharply. Holding $C_D A$ constant introduces a proportional scaling error in the retrieved density, which is largely absorbed by the multiplicative de-biasing of Eq. (2).

Table 2. Force modelling parameters and spacecraft characteristics used in this study.

Parameter	Value / Description	Reference
Third-body gravity	Sun & Moon point masses; DE421 ephemerides	(Folkner et al., 2009; Montenbruck & Gill, 2000)
Gravity field	EIGEN-6S4 100×100	(Förste et al., 2016)
Earth radiation	Knocke $1 \times 1^\circ$ model	(Knocke et al., 1988)
Solar radiation	Cannonball (C_r); cone shadow model	(Montenbruck & Gill, 2000)
Aerodynamic drag	Cannonball (C_D); density inverted-for	(Montenbruck & Gill, 2000)
Atmospheric winds	Co-rotating atmosphere (no winds)	(Montenbruck & Gill, 2000)
Reference frame	EME2000	(McCarthy, 1996)
Precession / nutation	IERS 2014 conventions	(McCarthy, 1996)
Polar motion & UT1	IERS C04 14	(Brzeziński et al., 2009)
Spacecraft	Mass (kg), Area (m^2), C_D , C_r	Reference
GRACE-FO-A	600.2, 1.04, 3.2, 1.5	(Mehta et al., 2013)
CHAMP	522.0, 1.00, 2.2, 1.0	(Mehta et al., 2017)

Bezděk (2010); Ray and Scheeres (2020) demonstrated that modelling the Earth’s gravity field to high degree and order is essential to prevent gravitational signal leakage into the drag residual. We model the gravity field to degree and order 100 (Table 2).

The drag acceleration is isolated as

$$\mathbf{a}_{\text{drag}} = \mathbf{a}_{\text{obs}} - \mathbf{a}_{\text{con}} - \mathbf{a}_{\text{srp}} - \mathbf{a}_{\text{erp}} \quad (4)$$

and projected onto the along-track direction defined by the atmospheric-relative velocity $\mathbf{v}_{\text{rel}} = \mathbf{v} - (\boldsymbol{\omega}_{\oplus} \times \mathbf{r})$. The instantaneous density at each timestep follows from the standard drag equation:

$$\rho(t_i) = \frac{2 a_{\text{drag},\parallel}(t_i) m}{C_D A v_{\text{rel}}^2(t_i)} \quad (5)$$

where the symbols are defined as follows:

- $\mathbf{a}_{\text{obs}}$: Total observed acceleration (from numerical differentiation)
- $\mathbf{a}_{\text{drag}}$: Acceleration due to drag (vector)
- $a_{\text{drag},\parallel}$: Scalar along-track component of the drag acceleration
- $\mathbf{v}_{\text{rel}}$: Satellite velocity relative to the co-rotating atmosphere
- v_{rel} : Magnitude of $\mathbf{v}_{\text{rel}}$
- $\mathbf{v}$: Satellite velocity in inertial space
- $\boldsymbol{\omega}_{\oplus}$: Angular rotation-rate vector of the Earth
- $\mathbf{r}$: Satellite position vector
- A : Cross-sectional area of the satellite
- ρ : Atmospheric density
- C_D : Drag coefficient
- m : Satellite mass

The pointwise density series is then integrated over a chosen arc using the Picone velocity weighting (Eq. (1)) to yield the arc-effective density. This integration step replaces the temporal smoothing filters used in previous numerical-differentiation studies. Crucially, the arc over which the density is extracted is independent of the span used for the cubic-spline interpolation and differentiation, which always uses the full multi-day SP3 ephemeris. The density-extraction arc is a sliding window that can range from a few minutes to a full orbital revolution, and its length controls the trade-off between temporal resolution and noise suppression.

2.3.3 Accelerometer-Derived Densities

The latest version of the TU Delft accelerometer-derived atmospheric densities were obtained for CHAMP and GRACE-FO-A (Siemes et al., 2023).

While accelerometer-derived densities are frequently regarded as a robust benchmark (Sutton et al., 2021; Mehta et al., 2022; Mutschler, Axelrad, et al., 2023), they are not free of systematic error. Aruliah et al. (2019) found that CHAMP-derived zonal winds were 1.5–2.0 times larger than co-located ground-based Fabry-Perot Interferometer observations, and Doornbos et al. (2010) showed that the choice of wind model in the density inversion can alter derived densities by up to 20%. Thus, while accelerometer data serves as the best available benchmark, its potential limitations should be kept in mind.

2.3.4 Energy Dissipation Rate Method

The Energy Dissipation Rate (EDR) method recovers atmospheric density from the change in orbital energy over an arc, avoiding the numerical differentiation of velocity that is central to POD-accelerometry. Under two-body dynamics the Jacobi energy of a satellite in the co-rotating Earth-Centred Earth-Fixed (ECEF) frame is conserved; departures from conservation are attributable to non-gravitational forces, principally atmospheric drag (Picone et al., 2005; King-Hele & Walker, 1987). EDR-based approaches underpin several operational density models (Storz, 2002; Bowman, 2003) and have been employed extensively in recent literature (Hejduk et al., 2013; Pilinski et al., 2016; Sutton et al., 2021).

This study follows the formulation of Sutton et al. (2021) and Ray et al. (2024). The Jacobi energy at each SP3 epoch is evaluated as

$$\xi = \frac{v_{\text{ECEF}}^2}{2} - \frac{\omega_{\oplus}^2(x^2 + y^2)}{2} - \frac{\mu}{r} - U_{\text{ns}} \quad (6)$$

where v_{ECEF} is the satellite speed in the ECEF frame, $\omega_{\oplus}$ is the Earth’s rotation rate, μ is the gravitational parameter, r is the geocentric distance, and U_{ns} is the non-spherical gravitational potential (EIGEN-6S4, degree and order 100). Because the gravitational field is conservative, the potential energy is a state function that depends only on position, not on the path taken. The Jacobi energy at any epoch can therefore be evaluated directly from the state vector at that epoch, and the energy change over an arc requires only two evaluations of the gravity field (at the endpoints) rather than a point-by-point summation along the entire trajectory. This makes the EDR method substantially cheaper per arc than POD-accelerometry, an advantage that becomes significant when scaling to the hundreds or thousands of tracked objects in LEO envisaged for next-generation density assimilation.

The energy change over an arc $[t_0, t_1]$ attributable to drag is obtained by subtracting the work done by the third-body (3BP) and radiation-pressure (SRP, ERP) perturbations:

$$\Delta E_{\text{drag}} = -(\xi_1 - \xi_0) + \int_{t_0}^{t_1} \mathbf{a}_{\text{3BP}} \cdot \mathbf{v}_{\text{rot}} dt + \int_{t_0}^{t_1} \mathbf{a}_{\text{SRP}} \cdot \mathbf{v}_{\text{rot}} dt \quad (7)$$

where $\mathbf{v}_{\text{rot}} = \mathbf{v} - (\boldsymbol{\omega}_{\oplus} \times \mathbf{r})$ is the velocity in the rotating frame. Following Ray et al. (2024), the work integrals for SRP and ERP are included because their omission introduces appreciable error when the drag signal weakens; their inclusion adds no significant computational burden. Both EDR and POD-A use the identical force model (Table 2): EIGEN-6S4 gravity truncated at degree and order 100, Sun and Moon third-body perturbations, cannonball SRP with conical shadow, and Knocke Earth radiation pressure.

The arc-effective density follows directly from equating the drag energy dissipation to the work integral of the drag force (Sutton et al., 2021):

$$\rho_{\text{eff}} = \frac{2 m \Delta E_{\text{drag}}}{C_D A \int_{t_0}^{t_1} v_{\text{rel}}^2 |\mathbf{v}| dt} \quad (8)$$

The denominator is precisely the Picone velocity-weighted integral of Eq. (1), so the EDR method produces the effective density by construction. No post-processing smoothing

or temporal averaging is required. As with POD-accelerometry, the arc $[t_0, t_1]$ can range from a few minutes to a full orbital revolution. Unlike POD-accelerometry, however, EDR uses the SP3 positions and velocities directly at their native 30 s cadence; there is no spline interpolation or differentiation step. The arc endpoints alone determine the gravitational energy change, with only the smaller third-body and radiation-pressure work integrals requiring numerical quadrature along the arc.

2.4 Selection of Evaluation Metrics

Thermospheric densities are approximately log-normally distributed (Emmert, 2015; Picone et al., 2005); evaluation metrics should therefore operate in log-space to treat multiplicative errors symmetrically. Following Fitzpatrick et al. (2025), we adopt the standard deviation of the log-ratio as the primary scatter metric:

$$\sigma = \text{std}\left\{\ln \rho_{\text{method}}(t_i) - \ln \rho_{\text{ACC}}(t_i)\right\}_{i=1}^N \quad (9)$$

which measures the dispersion of the retrieval around the reference after any systematic bias has been removed. For practical interpretation, σ is converted to a percentage scatter (Fitzpatrick et al., 2025):

$$\text{SD\%} = 100 (e^\sigma - 1) \quad (10)$$

Lower values indicate tighter agreement with the accelerometer reference.

In addition, we report the coefficient of determination (r^2) between the retrieved and reference density time series, following Ray et al. (2024):

$$r^2 = \left[\frac{\sum_i (\rho_{\text{method},i} - \bar{\rho}_{\text{method}})(\rho_{\text{ACC},i} - \bar{\rho}_{\text{ACC}})}{\sqrt{\sum_i (\rho_{\text{method},i} - \bar{\rho}_{\text{method}})^2 \sum_i (\rho_{\text{ACC},i} - \bar{\rho}_{\text{ACC}})^2}} \right]^2 \quad (11)$$

While σ quantifies the magnitude of relative errors, r^2 measures how faithfully the retrieval reproduces the timing and relative amplitude of density variations. Because r^2 is scale-invariant, it is unaffected by any residual multiplicative bias and directly quantifies the fraction of reference-signal variance explained by the retrieval. It is widely used in thermospheric model assessments (Bruinsma et al., 2021, 2018) and is well suited to this study because it cleanly separates the question of pattern fidelity from that of magnitude accuracy. Together, the two metrics provide a complete picture: a method may have low scatter (σ) yet miss the temporal pattern (low r^2), or vice versa.

Finally, we report the root-mean-square relative error:

$$\tilde{\rho}_{\text{RMS}} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{\rho_{\text{method}}(t_i) - \rho_{\text{ACC}}(t_i)}{\rho_{\text{ACC}}(t_i)} \right)^2} \quad (12)$$

This metric is identical to $\tilde{\rho}_{\text{RMS}}$ of Ray et al. (2024) (their Equation 3), who use it as their primary accuracy measure in their simulation-based comparison of POD-accelerometry and EDR. Reporting $\tilde{\rho}_{\text{RMS}}$ here therefore permits direct comparison with their predictions. Unlike σ and SD%, which operate in log-space, $\tilde{\rho}_{\text{RMS}}$ is computed in native (linear) density space and directly expresses the typical fractional error a user would encounter on any single density estimate. It is therefore the most immediately interpretable metric for operational applications such as conjunction screening, where the absolute magnitude of the density error matters more than its distributional properties. Because $\tilde{\rho}_{\text{RMS}}$ weights each point equally in linear space, it is more sensitive to large errors at high densities than the log-normal metrics; reporting both families of metrics therefore guards against conclusions that depend on the choice of error measure.

3 Results and Discussion

We evaluate both density retrieval methods across 49 geomagnetic storms (22 CHAMP, 27 GRACE-FO-A) using accelerometer-derived densities as the reference truth. All comparisons employ the metrics defined in Section 2.4. Results are organised around three findings: (i) both methods recover orbit-effective density with high fidelity, with EDR consistently outperforming POD-A; (ii) increasing the fit-span systematically improves precision; and (iii) sub-orbital arcs of 30–45 minutes recover temporal resolution while maintaining useful accuracy.

3.1 Orbit-Effective Validation

Figure 2 compares POD-A and EDR densities against accelerometer-derived densities across five configurations: orbit-effective densities at 1-, 2-, and 3-orbit fit-spans evaluated against matched-interval effective densities; single-orbit effective densities evaluated against native-cadence (10 s) accelerometer data; and sub-orbital arc-effective densities evaluated against native-cadence data. At orbit-effective cadence (~ 90 min), both methods agree closely with the accelerometer reference: POD-A achieves $r^2 = 0.970$ and $\text{SD}\% = 21.1\%$, while EDR achieves $r^2 = 0.978$ and $\text{SD}\% = 13.8\%$.

Although both methods agree closely with the accelerometer reference (POD-A: $r^2 = 0.970$, EDR: $r^2 = 0.978$), EDR outperforms POD-A on every metric at orbit-effective cadence and is roughly $6\times$ faster to run (Section 3.4), making it the preferred option when orbit-effective cadence is sufficient.

The scatter plots in Figure 2 exhibit a clear asymmetry: high densities are well captured by both methods, but as density decreases the spread about the 1:1 line increases and the retrievals progressively fail to track the accelerometer truth. This behaviour is visible across all panels but is most pronounced in the bottom row (sub-orbital, native reference), where the combination of shorter arcs and weaker drag signals amplifies the scatter at low densities. A related asymmetry is the greater density of points below the 1:1 line (bottom-right) than above it (top-left), indicating that the retrieval methods more frequently underestimate than overestimate. This is consistent with the temporal averaging inherent in both methods: storm-time density peaks are transient, and any finite arc or orbit average acts as a low-pass filter that preferentially suppresses these maxima.

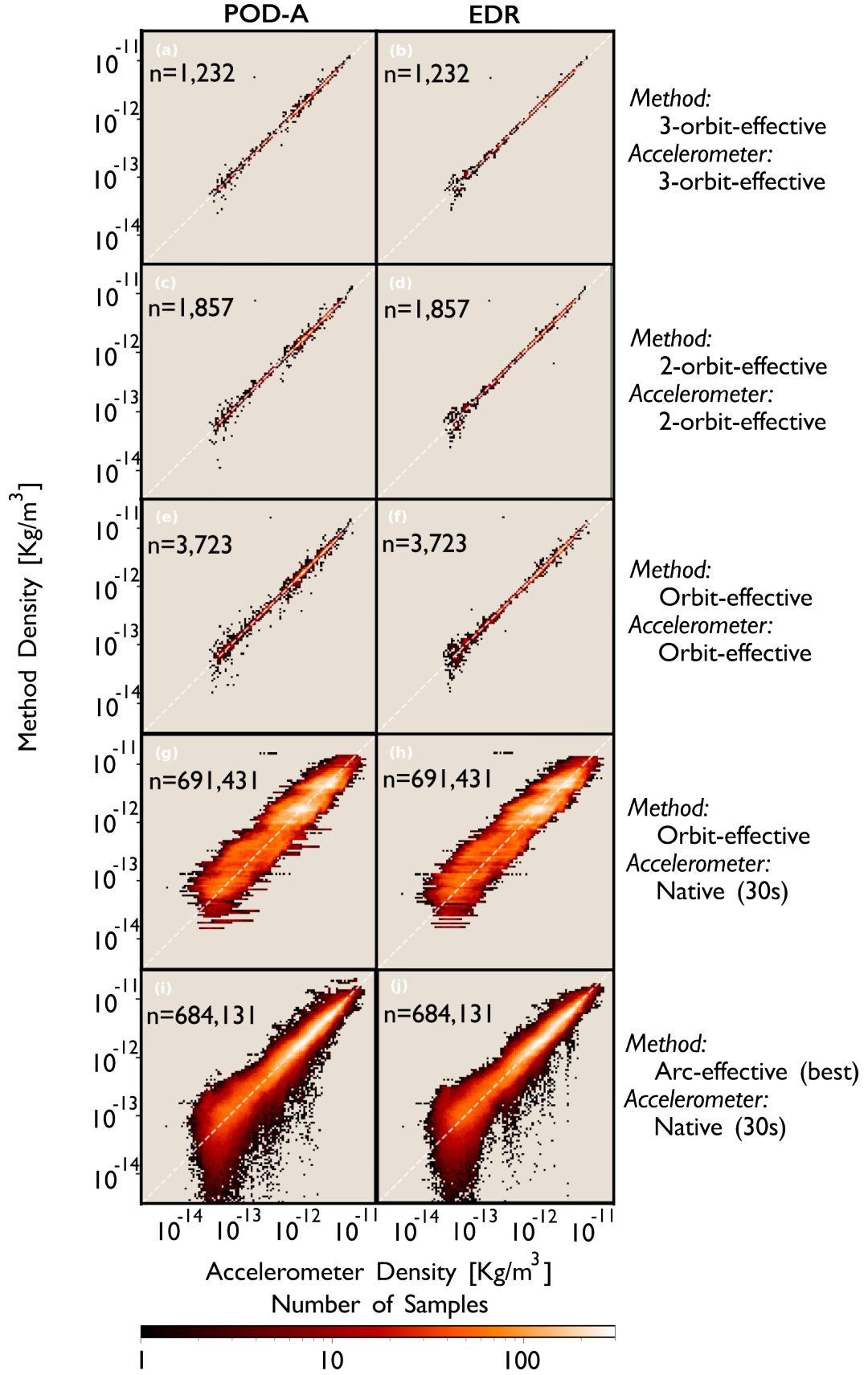


Figure 2. Scatter heat-maps of retrieved density versus accelerometer-derived density for POD-A (left) and EDR (right). Rows 1–3: orbit-effective densities at 3-, 2-, and 1-orbit cadence compared against matched-interval accelerometer effective densities. Row 4: 1-orbit effective density compared against native-cadence (10s) accelerometer data. Row 5: sub-orbital densities (fixed-optimal arc length) compared against native-cadence accelerometer data. Panel labels (a)–(j) correspond to the rows of Table 3. The dashed line denotes perfect agreement.

3.2 Dependence of Retrieval Error on Fit-Span and Drag Acceleration

Table 3 summarises performance across all five comparison configurations shown in Figure 2. Among the orbit-effective comparisons (top three rows), all metrics improve as the fit-span lengthens.

Table 3. Evaluation metrics (defined in Section 2.4) pooled across all 49 storms, as a function of fit-span and accelerometer reference cadence. “Matched” indicates that accelerometer-derived effective densities were computed over the same interval using the velocity weighting of Eq. (1); “native” indicates comparison against the 10 s accelerometer time series. The native-reference rows quantify the information lost to temporal averaging, not the accuracy of the retrieval method itself. The final three metrics report the in-storm per-orbit (or per-arc, for native rows) bias relative to ACC, after applying the pre-storm-calibrated debiasing factor β of Eq. (2): median, Q25 and Q75. Rows are listed in the same order as the panels in Figure 2.

Fit-span	ACC ref.	SD%		r^2		σ		$\tilde{\rho}_{\text{RMS}}\%$		median bias %		Q25 %		Q75 %	
		POD-A	EDR	POD-A	EDR	POD-A	EDR	POD-A	EDR	POD-A	EDR	POD-A	EDR	POD-A	EDR
3-orbit eff (~270 min)	matched	14.1	13.3	0.987	0.993	0.132	0.124	10.0	9.3	+0.3	+0.6	-1.8	-1.5	+2.2	+2.9
2-orbit eff (~180 min)	matched	15.3	13.9	0.984	0.988	0.143	0.131	11.0	10.1	+0.1	+0.7	-2.1	-1.8	+2.1	+3.0
1-orbit eff (~90 min)	matched	21.1	13.8	0.970	0.978	0.191	0.130	13.4	11.8	0.0	+0.7	-2.4	-2.3	+2.2	+3.6
1-orbit eff (~90 min)	native	59.6	56.0	0.794	0.800	0.467	0.445	55.0	55.1	+2.4	+7.4	-22.1	-18.3	+39.1	+46.8
Sub-orbital (best)	native	45.5	41.8	0.930	0.938	0.375	0.350	34.4	33.3	+0.3	+3.0	-14.7	-12.0	+20.4	+23.9

This improvement is expected: longer integration intervals average over more of the stochastic noise in both the retrieval and the reference. EDR outperforms POD-A on nearly all metrics and fit-spans, though the gap narrows at longer integration intervals. 95 % confidence intervals on the matched-reference metrics are exact under the log-normality of the residuals established in Section 2.4: the sample variance of normally distributed log-residuals follows a chi-squared distribution with $n - 1$ degrees of freedom (used here to interval-bound σ and $\tilde{\rho}_{\text{RMS}}^2$), and the Fisher z -transform provides the corresponding inferential interval for the Pearson correlation underlying r^2 . Applying these to the matched 1-orbit row, σ is indistinguishable between methods ($\sigma_{\text{POD-A}} = 0.115$ [0.112, 0.118] vs $\sigma_{\text{EDR}} = 0.116$ [0.113, 0.119]), whereas r^2 and $\tilde{\rho}_{\text{RMS}}$ differ significantly ($r^2 = 0.985$ [0.984, 0.986] vs 0.992 [0.991, 0.992]; $\tilde{\rho}_{\text{RMS}} = 13.5\%$ [13.2, 13.8] vs 12.0% [11.8, 12.3]). At single-orbit cadence the two methods produce similarly-scattered errors on a typical orbit (equivalent σ), but EDR follows the time-variation of the accelerometer truth more faithfully (higher r^2) and produces fewer catastrophic-error orbits (lower $\tilde{\rho}_{\text{RMS}}$, since $\tilde{\rho}_{\text{RMS}}$ squares the %-error and is dominated by the worst orbits). The jump from matched to native reference at 1-orbit cadence (rows 3–4) is substantial, quantifying the information lost to orbit averaging and motivating the sub-orbital analysis in Section 3.3. The sub-orbital row recovers $\sim 77\%$ of the lost r^2 .

Figure 3 disaggregates these pooled $\tilde{\rho}_{\text{RMS}}$ values by per-orbit mean drag acceleration and overlays the simulation-based predictions of Ray et al. (2024) (their Figure 13, 5 cm POD noise line). A direct comparison requires care: our results are orbit-effective densities (~ 90 min cadence), whereas Ray et al. (2024) report the best-resolution retrieval for each method and window combination; moreover, their POD-accelerometry implementation uses a Kalman-smoother filter, which involves substantially more processing than the numerical-differentiation approach used here (Section 2.3.2). The POD-A and EDR heatmaps broadly follow a similar shape, with both methods showing decreasing error at higher drag accelerations. The difference panel (right) reveals that EDR places slightly more orbits in lower-error bins, particularly at stronger drag accelerations ($> 10^{-7} \text{ m s}^{-2}$), visible as a concentration of blue (EDR-favoured) cells in the lower-right region of that panel. Our EDR median curve is similar in shape and magnitude to the Ray et al. (2024) prediction, while our POD-A median sits above their prediction across most of the drag range, consistent with the simpler numerical-differentiation approach

used here compared with the Kalman-smoother implementation modelled by Ray et al. (2024).

With these caveats, the broad trends are consistent. The distribution of per-orbit drag acceleration across our dataset is shown in Figure 4: CHAMP orbits are concentrated at 10^{-7} – 10^{-6} m s^{-2} , while GRACE-FO-A spans a broader and weaker range extending down to $\sim 10^{-8}$ m s^{-2} . At low drag (10^{-8} – 10^{-7} m s^{-2} , predominantly GRACE-FO-A orbits) $\tilde{\rho}_{\text{RMS}}$ rises to 10–25%, in line with the predicted sensitivity to drag signal strength. Below $\sim 10^{-8}$ m s^{-2} both methods exceed 30%, indicating a practical noise floor for orbit-effective retrieval. At these lower drag-acceleration levels, non-conservative forces such as SRP and Earth radiation pressure become comparable in magnitude to drag (Montenbruck & Gill, 2000); however, drag remains one of the most difficult perturbations to model owing to its highly variable nature, whereas SRP and ERP are often reasonably approximated by relatively simple models (e.g. once-per-revolution empirical corrections). At high drag ($> 10^{-7}$ m s^{-2} , predominantly CHAMP orbits) both methods achieve $\tilde{\rho}_{\text{RMS}} \approx 5$ –8%, with EDR slightly outperforming POD-A. At low drag, where force-model errors dominate over method-specific noise, both methods converge.

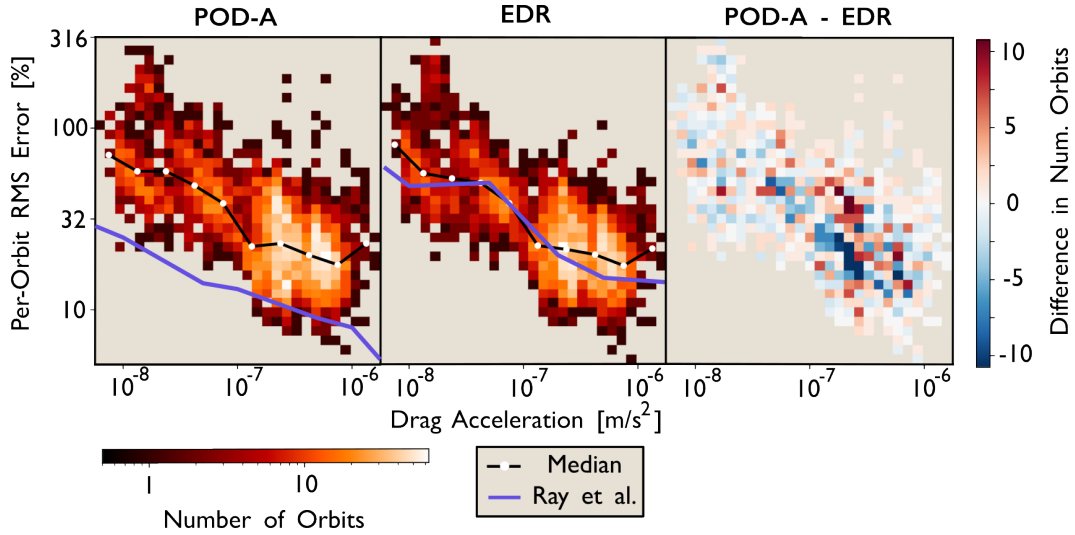


Figure 3. Heatmap of per-orbit RMS error ($\tilde{\rho}_{\text{RMS}}$) versus per-orbit mean drag acceleration for POD-A (left) and EDR (centre), pooled across all 49 storms. Colour intensity indicates the number of orbits in each bin. Dashed black lines show the binned median; solid coloured lines show the predictions of Ray et al. (2024) (their Figure 13, 5 cm POD noise line). The right panel shows the difference (POD-A – EDR): blue cells contain more EDR orbits, red cells contain more POD-A orbits.

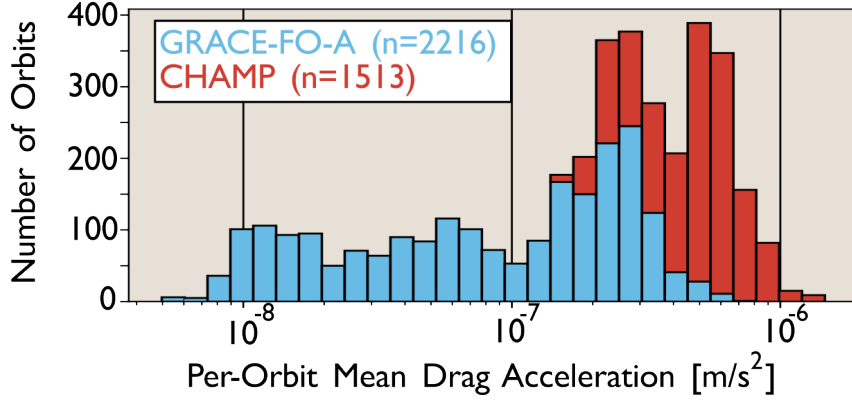


Figure 4. Distribution of per-orbit arithmetic-mean drag acceleration, computed from the ACC data, for CHAMP ($n = 1,513$ orbits) and GRACE-FO-A ($n = 2,211$ orbits) across all analysed storms.

3.3 Sub-Orbital Density Retrieval

Orbit-effective cadence (~ 90 min) is commonly used in density retrieval and assimilative studies (Matsuo et al., 2012; Sutton, 2018), in part because periodic perturbations such as SRP tend to cancel over a full revolution. However, storm-time thermospheric density varies on much shorter timescales. Oliveira et al. (2017) showed that high-latitude density responds within minutes of storm onset and that perturbations propagate to mid-latitudes in tens of minutes via travelling atmospheric disturbances, with global thermalisation taking approximately three hours. Zesta and Oliveira (2019) found that for extreme events the density perturbation reaches the equator within 1.5 hours of onset, shorter than a single orbital period. Bruinsma et al. (2021) found that model errors during storms reach 40–60%, with the largest degradation occurring during the rapid main-phase heating that orbit-averaging smooths over. Higher-cadence density retrievals from POD data could help constrain these sub-orbital features, provided that the retrieval precision remains adequate at shorter arc lengths.

While increasing the fit-span improves precision, it does so at the cost of temporal resolution. Row 4 of Figure 2 illustrates this directly: the same one-orbit effective densities from the previous sections, when compared against native-cadence (10 s) accelerometer data rather than matched effective densities, yield $r^2 = 0.80$ and $\text{SD}\% \approx 56$ –60%. This degradation does not reflect a failure of the retrieval method. Rather, it quantifies the density variability within a single orbit that orbit-averaging cannot resolve.

Sub-orbital arcs provide a route to recovering temporal resolution while retaining useful precision. To quantify this, each storm was processed at 20 arc lengths (2, 3, 5, 7, 10, 15, 20, 30, 45, 60, 75, 90, 100, 120, 135, 150, 180, 210, 240 and 270 min) for both POD-A and EDR. For each storm–satellite–method combination, the “best” arc length is the window that maximises the Pearson correlation against the native-cadence (10 s) accelerometer truth. Using these per-storm best windows, POD-A achieves $r^2 = 0.930$ and $\text{SD}\% = 45.5\%$; EDR achieves $r^2 = 0.938$ and $\text{SD}\% = 41.8\%$ (Figure 2, row 5). These represent a substantial improvement over the orbit-effective-vs-native comparison ($r^2 \approx 0.80$, $\text{SD}\% \approx 56$ –60%). Sub-orbital arcs thus recover real intra-orbit density structure that orbit averaging destroys.

Figure 5 shows how per-storm r^2 and σ vary with arc length. When evaluated against native-cadence accelerometer truth (blue), the per-storm median r^2 peaks at ~ 45 minutes (reaching ~ 0.76) then declines as over-smoothing erases real density structure. These

per-storm median values are notably lower than the pooled r^2 values in Table 3 ($r^2 = 0.93\text{--}0.94$). This discrepancy is an instance of ecological correlation (Robinson, 1950): pooling data across storms with different mean density levels introduces inter-storm variance that the pooled correlation can exploit. A method need only distinguish high-density storms from low-density storms to achieve a high pooled r^2 , even if it captures little of the intra-storm variability. The per-storm r^2 in Figure 5 isolates this intra-storm pattern fidelity and is therefore the more conservative measure of temporal resolution. Against resolution-matched truth (orange), both metrics improve with increasing window length because retrieval and reference are smoothed identically. The growing population of co-operatively tracked satellites operating within the thermosphere offers a potential route to mitigating this noise penalty: it may be that densities retrieved independently from many neighbouring satellites can be averaged to reduce scatter while preserving the higher temporal resolution. Studies quantifying the impact of commercial-grade POD noise on retrieval quality would be particularly valuable in this context.

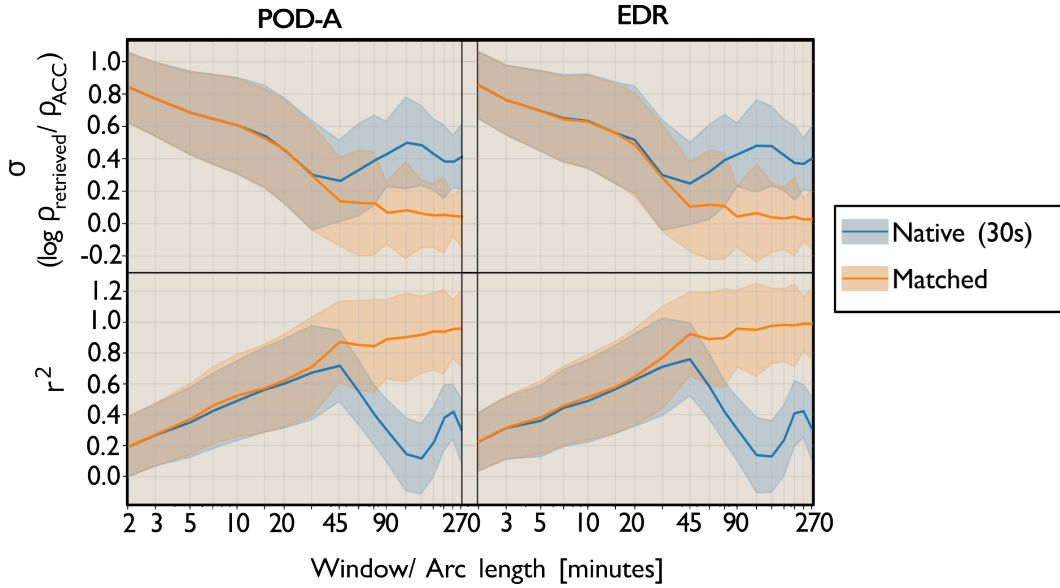


Figure 5. Per-storm σ (top) and r^2 (bottom) as a function of arc length, for POD-A (left) and EDR (right). Each storm yields one σ and one r^2 value at each arc length; solid lines show the median of these per-storm values across all storms, and shaded bands show $\pm 1\sigma$ of the inter-storm spread. Blue: evaluated against native-cadence (0.1 Hz) accelerometer truth. Orange: evaluated against resolution-matched (same-window-averaged) truth. Note that these per-storm median r^2 values are lower than the pooled r^2 in Table 3 because pooling across storms with different mean density levels introduces inter-storm variance that inflates the pooled correlation (see text).

These results can be placed in broader context by comparison with prior studies that validate orbit-derived densities against accelerometer references. Bernstein et al. (2021) report the same effective density quantity to assess Two-Line Element (TLE)-derived densities from CHAMP and GRACE against accelerometer effective densities. For their best TLE processing configuration (4-day moving window), they report Pearson correlations of $R = 0.956$ during solar maximum (2004) and $R = 0.794$ during solar minimum (2009), with the standard deviation of relative error ranging from approximately 3% at high drag to 10% at low drag for 8-day windows. Our POD-derived effective den-

sities at single-orbit cadence (~ 90 min) achieve $r^2 = 0.97\text{--}0.98$ with $\tilde{\rho}_{\text{RMS}}$ of 12–13%, representing improved accuracy at much finer temporal resolution. This improvement is consistent with the higher quality of cooperative POD data relative to the uncooperative TLE data used by Bernstein et al. (2021).

Other studies have validated POD-derived densities against accelerometer data using different comparison protocols. Calabia and Jin (2017) compared along-track densities derived from GRACE POD data directly against accelerometer measurements at each orbital position, reporting normalised root-mean-square deviations of 5–12% and a mean bias of 7%. Li et al. (2017) used an energy-balance approach with CHAMP POD data and found the standard deviation of the density ratio (POD to accelerometer) to be 0.21–0.33 at 20-minute cadence. These studies employ point-by-point or arithmetic-mean comparisons rather than the velocity-weighted effective density matching used here and in Bernstein et al. (2021), so direct numerical comparison of error metrics is not straightforward; nevertheless, the overall accuracy levels are broadly consistent with the results reported in this work.

3.4 Computational Run-time

Run-time tests were performed under Python 3.11 on an Apple M1 (10-core) processor equipped with 32 GB of unified memory. For each retrieval technique we bracketed the per-orbit inversion with the high-resolution wall-clock timer `time.perf_counter()` from Python’s `time` module, after all requisite input files (SP3 ephemerides, space-weather indices, etc.) had been pre-loaded into memory. Across 1000 orbits spanning both satellites, the mean execution time per orbit was 0.185 ± 0.003 s for POD-A and 0.031 ± 0.001 s for EDR, giving a roughly $6\times$ speed advantage for EDR. I/O and plotting were excluded so that the reported figures reflect only the computational cost of the retrieval algorithms themselves. Tests were executed serially (single OS thread).

4 Conclusions and Implications

This study presents the first multi-timescale evaluation of POD-derived thermospheric densities during geomagnetic storms, comparing POD-accelerometry and Energy Dissipation Rate retrievals against accelerometer-derived effective densities over 49 storms spanning CHAMP (22 storms) and GRACE-FO-A (27 storms). Returning to the three questions posed in the introduction:

Q1: How accurately do EDR and POD-A recover orbit-effective density, and does either method hold a systematic advantage? Given scientific-quality POD data, both methods recover orbit-effective density with high fidelity ($r^2 \geq 0.97$, $\text{SD}\% \leq 22\%$). EDR outperforms POD-A on nearly every metric and fit-span (Table 3) and is roughly $6\times$ cheaper to compute, making it the preferred method at orbit-effective cadence and for scaling to large populations of tracked objects.

Q2: What is the trade-off between temporal resolution and retrieval accuracy as arc length is varied? When retrieved densities are compared against accelerometer effective densities computed over the same arc length, $\text{SD}\%$ improves with increasing fit-span, reaching $\sim 13\%$ at three orbits. However, this comes at the cost of temporal resolution: longer arcs smooth out real density variations, and per-storm median r^2 against native-cadence accelerometer data drops sharply beyond ~ 45 minutes (from ~ 0.76 at 45 min to ~ 0.31 at one orbit and ~ 0.12 at 135 min; Figure 5). Sub-orbital arcs of 30–45 minutes therefore represent the best compromise, achieving pooled $r^2 = 0.93\text{--}0.94$ against native-cadence data, though at the cost of increased uncertainty ($\text{SD}\% \approx 42\text{--}46\%$ vs $\sim 14\text{--}21\%$ at orbit-effective cadence).

Q3: What is the relationship between retrieval accuracy and drag-acceleration magnitude, and do the observed trends corroborate predictions in the literature? Retrieval accuracy depends strongly on drag signal strength: at per-orbit mean drag accelerations above $\sim 4 \times 10^{-7} \text{ m s}^{-2}$, $\tilde{\rho}_{\text{RMS}}$ is roughly 10% or less, but below this threshold errors increase rapidly, exceeding 30% as drag approaches $\sim 10^{-8} \text{ m s}^{-2}$. These trends are broadly consistent with the simulation-based predictions of Ray et al. (2024). Our EDR results are slightly better than their predictions but otherwise in good agreement. Our POD-A results are slightly worse, which is expected given that Ray et al. (2024) model a more involved Kalman-smoother-based POD-accelerometry implementation, whereas this study uses a simpler numerical-differentiation approach (Section 2.3.2).

For current operational assimilative schemes, orbit-effective densities are the recommended product in view of their higher precision. However, as next-generation models mature and cooperative POD data from emerging LEO constellations become available, the higher correlation with native-cadence accelerometer data offered by ~ 45 minute arcs represents an attractive pathway to increased spatio-temporal resolution. The increased uncertainty at this sub-orbital cadence may be attenuated by fusing data from many neighbouring satellites simultaneously.

More broadly, the growing pool of cooperative POD data from scientific and commercial missions offers a dataset of opportunity for thermospheric monitoring (Fitzpatrick et al., 2025; Arnold et al., 2023). Even at the single-satellite level, both retrieval methods deliver orbit-effective densities that are highly consistent with independent accelerometer measurements. Harnessing cooperative data from a fraction of today's $> 11,000$ LEO satellites could supply a dense, low-latency observational layer that informs thermospheric modelling precisely when density errors are most operationally consequential: during geomagnetic storms (Parker & Linares, 2024; Parker et al., 2024; Berger et al., 2023).

5 Future Work

This analysis uses two near-polar, near-circular spacecraft; extension to diverse orbital geometries and to lower-quality commercial POD data (Arnold et al., 2023; Fitzpatrick et al., 2024) is a natural next step (Table 1), though such platforms generally lack accelerometers for independent validation. Lower drag-acceleration levels ($\lesssim 10^{-7} \text{ m s}^{-2}$) remains sparsely sampled and warrants dedicated study at higher altitudes or during quiet times (Ray et al., 2024). Finally, an adaptive windowing scheme that shortens arcs during high-drag storm peaks and lengthens them during quieter intervals could improve sub-orbital temporal resolution precisely when it matters most.

Open Research Section

Accelerometer-derived thermospheric densities for CHAMP and GRACE-FO are openly available from the TU Delft database (Siemes et al., 2023) at <https://thermosphere.tudelft.nl/>. Precise orbit ephemerides for GRACE-FO and CHAMP were obtained from the GFZ ISDC archive (Schreiner et al., 2022) at <ftp://isdctftp.gfz-potsdam.de/>. Geomagnetic and solar drivers were obtained from NOAA SWPC (<https://www.swpc.noaa.gov/>) and the Space Environment Technologies portal (<https://sol.spacenvironment.net/jb2008/>). The analysis software and pre-computed intermediate data used in this study are archived at Zenodo (Constant et al., 2026) under a Creative Commons Attribution 4.0 International licence. The repository contains all data and scripts needed to reproduce all figures and tables in this paper; source code is also available at <https://github.com/CharlesPlusC/storm-density-edr-vs-pod>.

Author Contributions

Charles Constant: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Visualization, Writing – Original Draft, Writing – Review & Editing. **Indigo Brownhall:** Methodology, Software, Validation, Writing – Review & Editing. **Anasuya Aruliah:** Conceptualization, Supervision, Writing – Review & Editing. **Marek Ziebart:** Conceptualization, Supervision, Writing – Review & Editing. **Santosh Bhattarai:** Conceptualization, Supervision, Writing – Review & Editing.

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